OPENWORLDX · MIRA

Product & Architecture Overview

A WhatsApp career assistant for campus talent, and a recruiter portal that matches job descriptions to candidates and connects them through Mira.

Version 1.0 · 19 October 2026 · Firebase project `openworld-8f410` · openworldx.com

1. What the product does
2. Candidate flow
3. Recruiter (HR) flow
4. Intro lifecycle
5. Critical user journeys
6. Technology we use
7. Architecture & data model
8. How it scales
9. Limitations & known risks
10. WhatsApp templates
11. Configuration
12. Recommended next steps

1. What the product does

For candidates (WhatsApp)

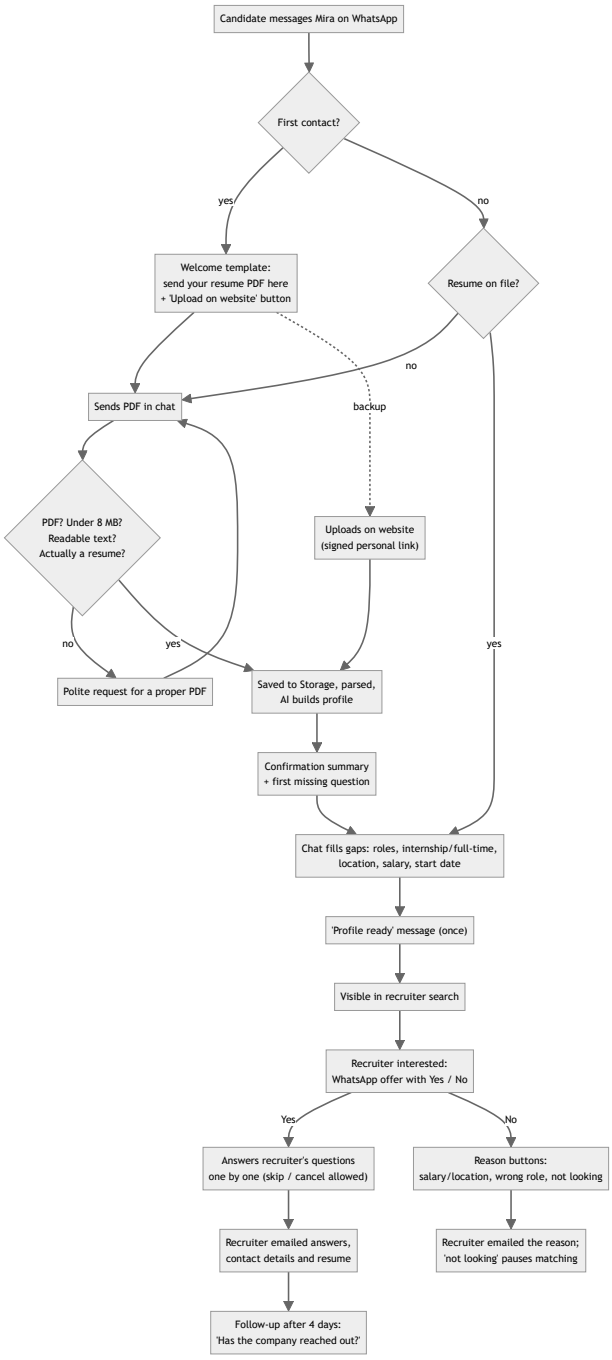
- Message Mira on WhatsApp; send the resume as a PDF in the chat (or upload on the website).
- Mira reads it, builds a profile, confirms what she picked up, and asks for missing details: target roles, internship or full-time, location, expected salary, start date.
- When a recruiter is interested, Mira asks the candidate Yes/No, collects the recruiter's questions, and sends the answers.
- Follow-ups, reminders and "still looking?" check-ins keep profiles fresh.

For recruiters (openworldx.com/hire)

- Upload or paste a job description.
- AI ranks up to 10 matching candidates with a score, reasons and gaps.
- Full profile, resume and contact details on each card.
- "Reach out via Mira": a note plus free-text questions; Mira handles the conversation and emails the answers.
- A private results link shows live status for every candidate.

Privacy model (current decision): recruiters see the candidate's name, phone, email and resume *before* the candidate agrees. Mira's welcome template and replies tell candidates this. Every search records who ran it (company, name, email, IP).

2. Candidate flow

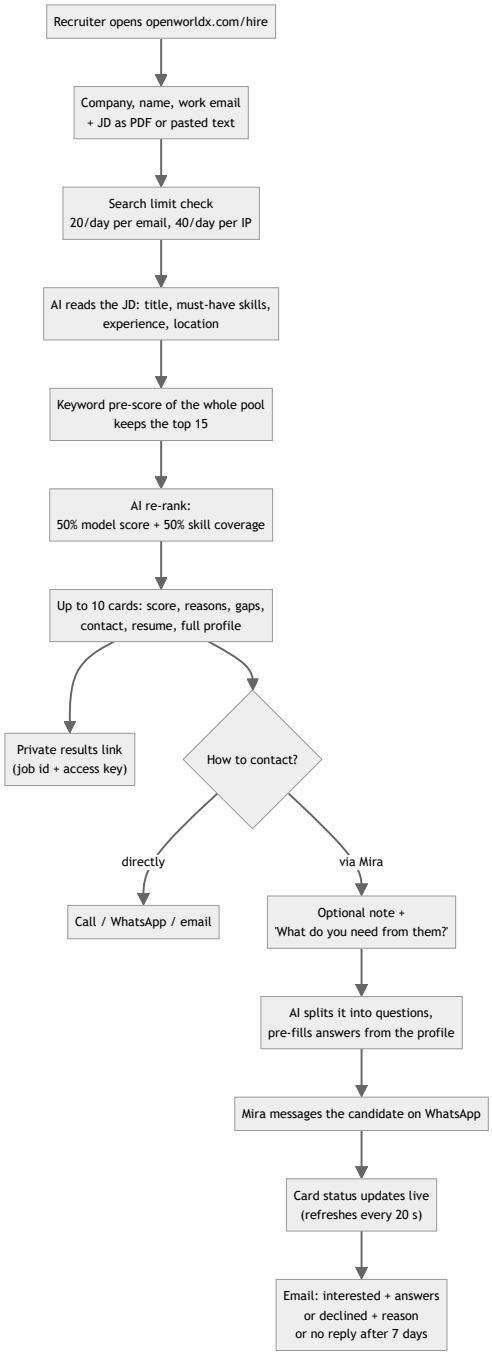


Candidate journey from first message to recruiter introduction.

Behaviour rules

- Once a resume is on file, Mira never asks for it again. A new PDF at any time counts as an update ("Updated your profile…").
- The upload link is shared only when the candidate asks for it, is on a laptop, or can't send the file in chat.
- Every message Mira sends is saved to chat history (last 40 messages), so later replies have full context.

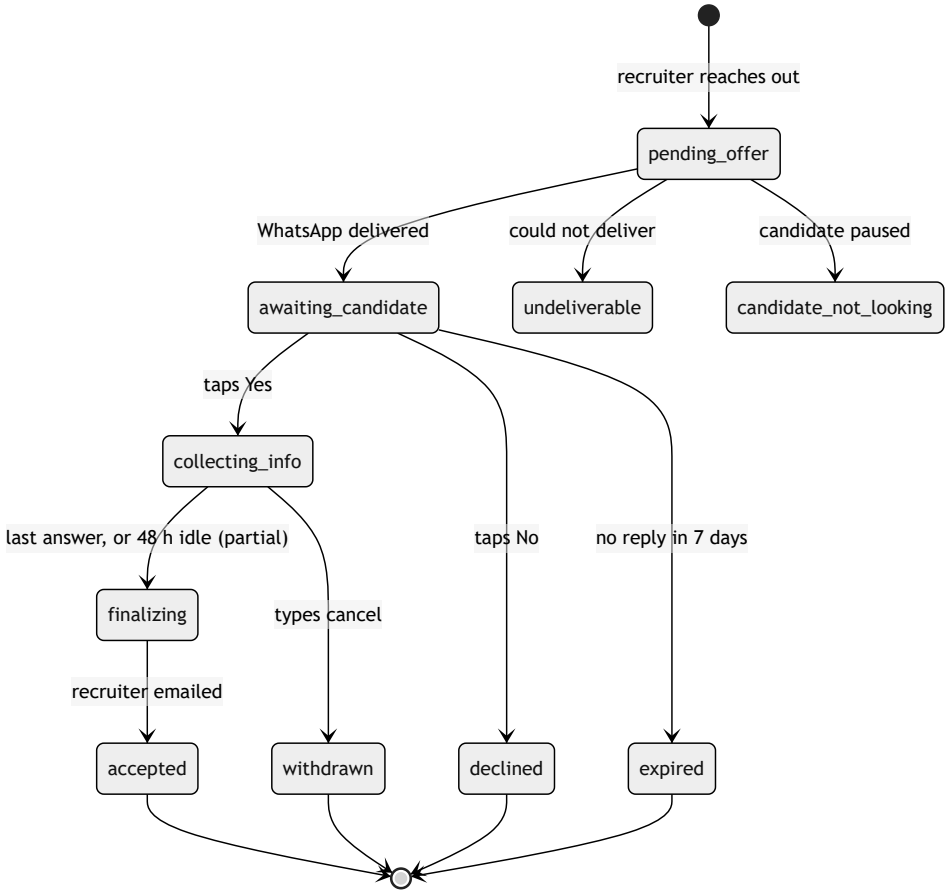
3. Recruiter (HR) flow



Recruiter journey from job description to candidate answers.

4. Intro lifecycle

Every "Reach out via Mira" creates one intro record. Status changes are done in database transactions, so a candidate's reply and the scheduler can never act on the same intro twice.



States stored in `hiringIntroRequests/{jobId}_{matchId}`.

Automatic jobs (every 2 hours, 8am–10pm IST)

Situation	After	What happens
Offer not answered	1 day	One reminder to the candidate (template if outside 24 h)
Offer still not answered	7 days	Expired; recruiter emailed the direct contact details
Stopped mid-questions	~20 hours	Reminder; template button "Continue" resumes the questions
Still stopped	48 hours	Recruiter emailed the partial answers
Declined, no reason tapped	3 hours	Decline email sent without a reason
Interested	4 days	"Has the company reached out?"; "Not yet" emails the recruiter a nudge
Candidate silent	45 days	"Still looking?" check-in; "Not right now" pauses matching
Email send stuck	10 minutes	Retried automatically

5. Critical user journeys

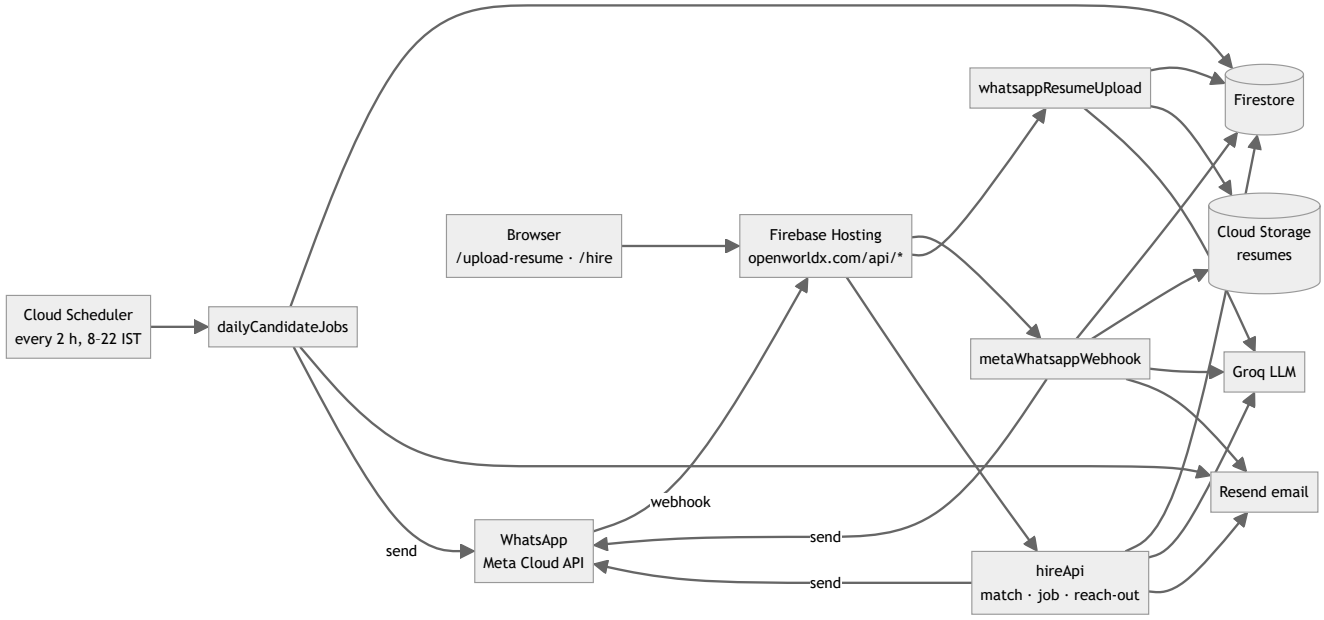
These journeys must work for the product to deliver value. Each lists how it succeeds and what happens when something goes wrong.

Journey	Steps	Success looks like	Failure handling
C1. First contact & resume in chat	Says hi → welcome template → sends PDF → "Reading it now…" → summary + first question	Profile built in under 30 s; no second request for the resume	Photo, Word, >8 MB, scanned or non-resume files get a specific fix-it message; duplicate deliveries from Meta are ignored
C2. Resume on the website	Taps "Upload on website" → preview → upload → redirected back to WhatsApp → confirmation arrives	Upload, parse and confirmation in one flow	Invalid/old link shows "get a new link"; parse failure triggers a WhatsApp message asking for a PDF
C3. Complete the profile	Mira asks one missing detail at a time → "Profile ready"	All 5 gap fields filled; ready message sent once	Candidate can correct anything in plain language
C4. Say yes to a recruiter	Offer → Yes → questions (with pre-filled confirmations) → "Done ✅"	Recruiter receives every answer by email	skip / cancel; questions about the role are answered or passed on; reminder + partial send if they go quiet
C5. Say no	Offer → No → reason button	Recruiter informed; matching paused if "not looking"	No reason in 3 h → email sent anyway
C6. Update or pause	Sends a new PDF / says they got a job	Profile refreshed / removed from search	45-day check-in catches stale profiles
R1. Find candidates	Fill company, name, email + JD → results in under a minute	Relevant, stable scores with clear reasons	Short JD rejected; unreadable PDF error; daily limits
R2. Reach out via Mira	Note + questions → candidate asked on WhatsApp → live status → email with answers	Answers in the inbox; reply goes to the candidate	Undeliverable status if WhatsApp fails; no-reply email after 7 days
R3. Come back later	Opens the saved results link	Same results with current statuses	Wrong or edited key → "link invalid"

6. Technology we use

Layer	Technology	Used for
Messaging	WhatsApp Business Cloud API (Meta Graph API v22)	Inbound webhook, text, reply buttons, templates, document download, read receipts
Backend	Firebase Cloud Functions 2nd gen, Node.js 22, ES modules, region asia-south1 (Mumbai)	Webhook, resume upload API, recruiter API, scheduled jobs
Scheduling	Cloud Scheduler (via Firebase `onSchedule`)	Reminders, expiries, follow-ups, check-ins
Database	Cloud Firestore	Candidates, resumes, chats, jobs, intros, rate limits, config
File storage	Cloud Storage for Firebase	Resume PDFs (`whatsapp-resumes/{phone}/…`) with download-token URLs
AI	Groq API, model `openai/gpt-oss-120b`, via Vercel AI SDK	Mira's replies, resume → profile, chat → profile updates, JD parsing, ranking, question splitting, answer reading, resume check
PDF parsing	`pdf-parse` 1.1.1 (server), `pdfjs-dist` 3.11 (browser preview)	Resume and JD text; first-page preview on the upload page
Email	Resend REST API	Recruiter emails: interested, declined, no reply, nudge
Frontend	React + Vite, Chakra UI, lucide-react icons, React Router	Landing page, `/upload-resume`, `/hire`
Hosting	Firebase Hosting + rewrites	`/api/whatsapp/meta`, `/api/upload-resume`, `/api/hire/**` → functions; SPA for everything else
Security	HMAC-signed upload links, random access keys per job, Meta webhook signature check (when the app secret is set), Firestore transactions	Stops link tampering, results scraping, fake webhooks and double processing

7. Architecture & data model



Every request goes through openworldx.com; four functions do the work.

Firestore collections

Collection	Contents
`whatsappCandidates/{phone}`	Profile: name, education, skills, roles, experience, preferences, summary, job-search status, last message time, profile-ready time
`whatsappResumes/{phone}`	Storage path, download URL, file name, source (web/WhatsApp), upload count, extracted text, processing status
`whatsappChats/{phone}`	Last 40 messages (user and Mira)
`whatsappProcessedMessages/{id}`	Message IDs already handled, to ignore Meta's duplicate deliveries
`hiringJobs/{id}`	Recruiter details, JD text, parsed JD, matches, access key, requester IP and browser
`hiringIntroRequests/{jobId}_{matchId}`	Intro status, note, questions and answers, candidate's questions, decline reason, delivery and email results, reminder times
`hireRateLimits/{hash}`	Daily search counters per email and per IP
`whatsappConfig/meta`	Real phone number ID and WhatsApp Business account ID, learned from webhooks

8. How it scales

What scales on its own

- **Functions:** autoscale. Each is capped at 10 instances × 80 concurrent requests, about 800 requests in flight per function. The cap is a single setting (`maxInstances`).
- **Firestore and Storage:** managed and effectively unlimited for this workload. Each candidate has their own documents, so there are no shared "hot" documents.
- **Stateless functions:** all state is in Firestore, so any instance can handle any message.

Bottlenecks, roughly in the order they'll be hit

Bottleneck	Why	Fix when needed
WhatsApp messaging limits	Meta caps business-initiated conversations per 24 h by tier (starts low, then rises through 1K → 10K → 100K as quality stays high). Marketing templates also have per-user frequency caps.	Business verification, keep quality high, prefer Utility wording, avoid spammy offers
Groq rate limits and latency	Each chat message makes 2 AI calls (reply + profile update); resumes, searches and answers add more. Plan limits on requests and tokens per minute cap throughput.	Paid tier, a smaller model for profile updates, retries with backoff, a queue
Recruiter search reads the whole pool	Every search loads all candidate profiles and keyword-scores them. Fine to a few thousand, then slow and costly.	Embeddings + Firestore vector search (or a search index) to pre-filter the top ~100
Work done inside the webhook	The webhook replies to Meta only after the AI work (5–20 s). Duplicate deliveries are already ignored, but heavy load means more retries.	Return 200 immediately; process via Cloud Tasks or Pub/Sub
Scheduler scans	The check-in job reads every candidate every 2 hours.	Query by indexed `lastInboundAt` ; split jobs into batches
Email volume	Resend's free plan allows about 100 emails a day.	Paid Resend plan

Approximate capacity today

With current settings, the system comfortably handles **hundreds to a few thousand active candidates** and dozens of recruiter searches a day. The first real limits will be the WhatsApp tier and the Groq plan, not Firebase.

9. Limitations & known risks

Must fix before going live

- **Email domain not verified in Resend.** Emails only reach the Resend account owner (akhileshthota30@gmail.com) until openworldx.com's DNS records are added.
- **Webhook signature check is off** until `META_APP_SECRET` is set. Until then, anyone with the URL can fake candidate replies.
- **Test mode allowlist.** `ALERT_ALLOWLIST` restricts proactive messages to one number; other candidates show as "couldn't reach".

Product & privacy

- Recruiter emails are not verified; anyone can type any company and email and see contact details. Only daily limits and the search record protect against scraping. Email verification (one-time code) is recommended once Resend is live.
- Showing contacts before consent needs clear disclosure to candidates (done in the welcome template and replies) and a privacy policy that covers India's data protection law, the DPDP Act.
- English only.

Technical

- Resumes in chat: PDF only. Photos, scanned PDFs and Word files are rejected with guidance. On the website, Word files are accepted but read poorly (treated as plain text).
- AI output varies run to run. Match scores are stabilised by blending in a fixed skill-coverage score, but rankings can still shift slightly.
- Matching uses keywords first, so candidates who describe skills with unusual words can be missed before the AI re-rank.
- Profile lists only grow. A skill can't be removed through chat (corrections to single fields work).
- Reminders run every 2 hours between 8am and 10pm IST, so they can arrive up to about 2 hours late.
- Outside WhatsApp's 24-hour window only approved templates can be sent. Marketing templates cost more and are frequency-capped by Meta.
- No automated tests, CI, or admin dashboard yet. Test data has to be cleaned manually.
- Chat history keeps the last 40 messages per candidate.

10. WhatsApp templates

All templates are approved, in English (`en`), on the WhatsApp Business account. Meta moved three of them from Utility to Marketing.

Template	Category	Body	Buttons	Used when
`openworldx_mira_welcome_v2` current welcome	Marketing	Header: Mira Hi {{1}}, I'm Mira from OpenWorldX. I help you find jobs that fit you. Step 1: Send your resume as a PDF right here in this chat (tap 📎 → Document). Step 2: Chat with me so I understand your skills and goals. Recruiters hiring on OpenWorldX can see your profile, resume and contact details. Message me anytime to pause. Footer: OpenWorldX	URL: "Upload on website" → `/upload-resume{{1}}` (signed personal link)	First message from a new candidate
`openworldx_recruiter_interest`	Marketing	Hi {{1}}, a recruiter is interested in you for the {{2}} role at {{3}}. Are you interested? If yes, I'll ask you a few quick questions from them and send your answers along with your resume.	Quick replies: "Yes, interested" · "No thanks"	Recruiter reaches out and the candidate is outside the 24 h window
`openworldx_questions_reminder`	Utility	Hi {{1}}, the recruiter from {{2}} is still waiting for a few answers from you about the {{3}} role. Tap below to continue whenever you're ready.	Quick reply: "Continue"	Candidate said yes but stopped answering for about 20 h
`openworldx_intro_followup`	Marketing	Hi {{1}}, quick check-in from Mira. Has {{2}} reached out to you about the {{3}} role yet?	Quick replies: "Yes, in touch" · "Not yet"	4 days after the candidate said yes
`openworldx_status_checkin`	Marketing	Hi {{1}}, it's Mira from OpenWorldX. Are you still looking for a new role? Let me know so I only send you jobs when you want them.	Quick replies: "Still looking" · "Not right now"	After 45 days with no message from the candidate

`openworldx_mira_start` previous welcome	**Marketing**	Hi {{1}}, I'm Mira. I help you find jobs that fit you. Step 1: Upload your resume (PDF) using the button below. Step 2: Chat with me here so I understand your skills and goals. Then I'll suggest the right jobs for you.	URL: "Upload Resume"	Not in use (kept as a fallback)

Variables {{1}}, {{2}}, {{3}} are filled with the candidate's first name, then the role or company in the order shown. Quick-reply buttons carry hidden IDs (for example `intro:yes:<id>`) that tell Mira which intro the tap belongs to. Inside the 24-hour window, Mira sends the same content as normal messages with buttons instead of templates.

11. Configuration

Settings live in `functions/.env`, which Git ignores. The values are secret and are not reproduced here.

Setting	Purpose
`META_ACCESS_TOKEN`, `META_PHONE_NUMBER_ID`, `META_WABA_ID`, `META_VERIFY_TOKEN`, `META_GRAPH_VERSION`	WhatsApp Cloud API access
`META_APP_SECRET` *(to add)*	Verifies that webhooks really come from Meta
`META_RESUME_TEMPLATE_NAME`, `META_RESUME_TEMPLATE_LANG`, `META_WELCOME_CHAT_HINT`	Welcome template
`META_MATCH_TEMPLATE_NAME`, `META_QUESTIONS_REMINDER_TEMPLATE_NAME`, `META_FOLLOWUP_TEMPLATE_NAME`, `META_CHECKIN_TEMPLATE_NAME`	Templates for messages outside the 24-hour window
`ALERT_ALLOWLIST`	Test mode: proactive messages only go to these numbers. Empty means everyone.
`AUTO_OFFER_STRONG_MATCHES`	Off by default. When on, candidates scoring 75 or more are messaged automatically on each new search.
`GROQ_API_KEY`	AI model access
`RESEND_API_KEY`, `RESEND_FROM`	Recruiter emails
`UPLOAD_LINK_SECRET`, `PUBLIC_APP_URL`	Signs personal upload links

12. Recommended next steps

#	Step	Why
1	Verify openworldx.com in Resend (3 DNS records); send from `mira@openworldx.com`	Recruiter emails are the main deliverable of the HR flow
2	Add `META_APP_SECRET`	Blocks fake webhook calls
3	Clear `ALERT_ALLOWLIST` to go live	Lets Mira reach every candidate
4	Recruiter email verification (one-time code)	Contact details are visible; stops anonymous scraping
5	Privacy policy and terms covering recruiter visibility (DPDP Act)	Legal clarity for candidates and recruiters
6	Move webhook work to a queue (Cloud Tasks)	Faster replies to Meta and resilience under load
7	Embeddings + vector search for matching	Searches stay fast past a few thousand candidates
8	Admin dashboard (candidates, jobs, intros, emails) and automated tests	Operations, debugging and safe changes
9	Word and scanned-PDF support (conversion and text recognition)	Fewer rejected resumes